



## Haemopoietic Disorders

### Learning Objectives

- ⇒ To know about brief introduction about haemopoietic diseases
- ⇒ To know about the clinical features of anaemia, nutritional anaemia, thalassemia, sickle cell anaemia, leukemia, thrombocytopenia
- ⇒ To know about the necessary physical examinations to be performed in above said diseases
- ⇒ To know about the necessary investigations to be ordered in the above said diseases and methods of interpretation

Anemia (see Figure 60) is a disease of multiple causations and its scientific management requires an exact classification which can be done only after a complete study of blood. There is a tendency among the health care professionals to analyse Hb analysis for the evaluation of anemia and they prescribe iron supplementation. Hb estimation merely detects the presence and severity of anemia. However, further tests like haemogram (CBC), peripheral smear study and biochemical investigations are required to classify anemia and appropriate management. In any case of anemia, therefore, it is not wise to begin treatment without CBC and peripheral smear study.

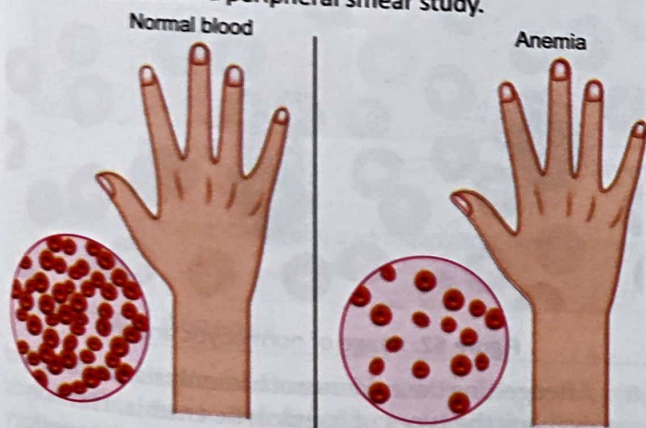


Figure 60: Image of normal and anemic patient

Among the different parameters observed in complete blood count (CBC), following parameters are of much clinical significance as far as the diagnosis of anemia is concerned.

- **Haemoglobin** – general indicator of anemia and polycythemia

- **MCV (Mean Corpuscular Volume)** – key parameter for classification of anemia
- **MCHC (Mean Corpuscular Haemoglobin Concentration)** – helps in the classification of anemia
- **RDW (Red cell Distribution Width)** – helps in differential diagnosis of anemia
- **RBC count** – may be either normal or decreased in anemia.

First step in the evaluation of anemia is careful observation of haemogram report along with peripheral study. Here the clinician must focus on MCV which helps in the morphological classification of anemia that is microcytic, normocytic and macrocytic anemia.

### Microcytic anemia

If MCV is decreased and peripheral smear study also shows microcytic picture of red blood cells clinician may think in the direction of microcytic anemia. The three major diagnostic possibilities for microcytic anemia are iron deficiency anemia, thalassemia and anemia of chronic disease (see Figure 61).

### Iron deficiency anemia (IDA)

Iron deficiency is the most common cause of microcytic anemia, that is the reason, when you get the microcytic picture of anemia, we have make the diagnosis of iron deficiency as the cause of anemia. Increased RDW (red cell distribution width) and thrombocytosis are diagnostic clues for iron deficiency anemia (IDA) in haemogram. Anisocytosis, poikilocytosis (variations in the shape of



RBCs), elliptocytosis are observed in peripheral blood smear. If haemogram and peripheral blood smear are directing towards IDA, then confirm IDA by asking the biochemical test. The important biochemical test to be done is serum ferritin level. Low level of serum ferritin is diagnostic of IDA. Other iron panel tests which include serum iron, transferrin-iron saturation % are decreased and TIBC (total iron binding capacity) and serum transferrin levels are increased. Bone marrow biopsy is not necessary for the diagnosis of IDA.

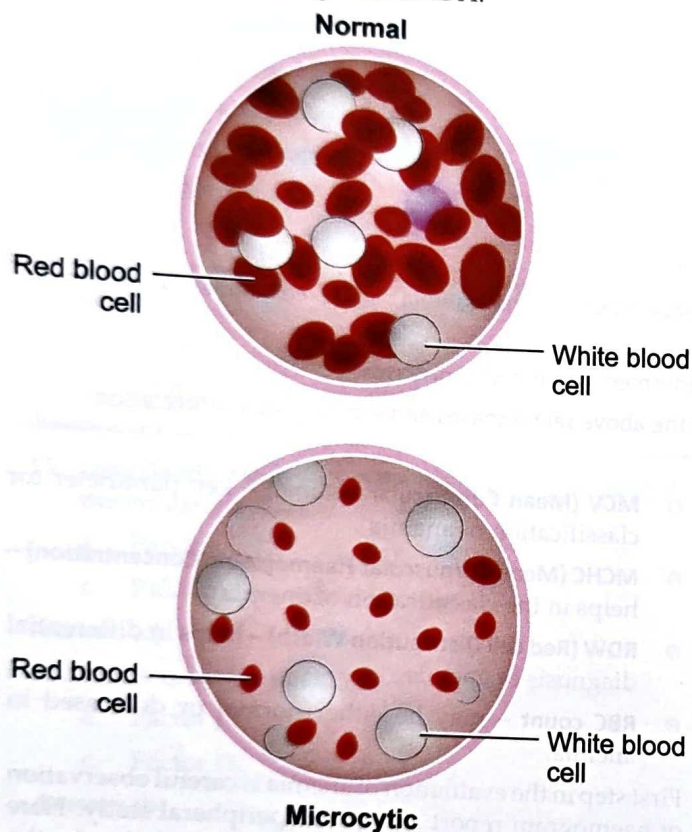


Figure 61: Image of normal and microcytic blood

#### Thalassemia

This is explained separately in this chapter elsewhere.

#### Anemia of chronic disease (ACD)

Rheumatoid arthritis, tuberculosis, temporal arteritis, chronic inflammation and infection are the usual cause of ACD. Renal cell carcinoma, Hodgkin lymphoma, castleman disease, myelofibrosis are the unusual causes of ACD. In this type of microcytic anemia, variations may be observed in total count, differential count, platelet count. Abnormal cells, blast cells may be reported in peripheral blood smear (PBS) in case of blood malignancy. In ACD, serum iron, TIBC, transferrin-iron saturation % and transferrin are decreased.

#### Normocytic anemia

In this type of anemia (see Figure 62), MCV is normal and

peripheral blood smear (PBS) shows normal sized RBCs. While evaluating normocytic anemia, clinicians may think of bleeding, nutritional anemia, anemia associated with renal insufficiency and haemolytic anemia.

- ⊗ If bleeding is implicated as the cause of anemia, then stool occult blood test, urine test can be done if indicated.
- ⊗ In case of nutritional anemia, iron and vitamin B12 or folic acid deficiencies can be ordered despite their usual association with microcytic and macrocytic anemia respectively. In these cases, RDW is usually increased and PBS report shows anisocytosis and dimorphic picture of RBC.
- ⊗ Anemia of renal insufficiency can be identified by checking serum creatinine level.
- ⊗ Another possibility to consider in normocytic anemia is haemolytic anemia. Thrombocytosis is usually observed and RDW is usually normal or elevated. Polychromasia, spherocytes, schistocytes may be reported in smear examination. Evaluation of haemolytic anemia begins by asking tests which give evidence of haemolysis. Tests in this line are LDH (lactate dehydrogenase), unconjugated bilirubin, serum haptoglobin. LDH and unconjugated bilirubin are elevated and haptoglobin is decreased in case of haemolysis. In urine, urobilinogen is raised, but there is not bilirubinuria. Clinicians can also get the evidence of intravascular haemolysis in the form of haemoglobinuria.

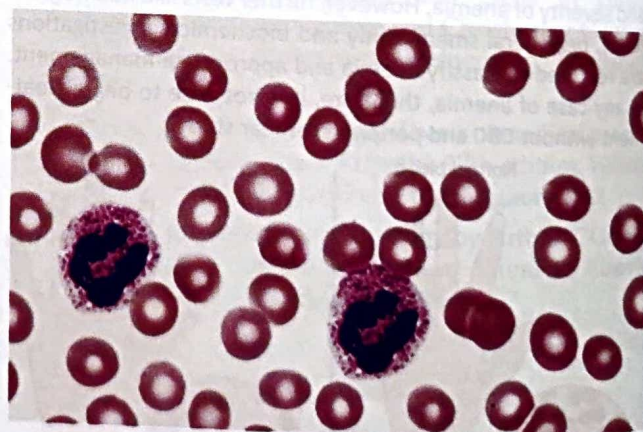


Figure 62: Image of normocytic anemia

- ⊗ After getting the evidence of haemolysis, next step is to evaluate the types of haemolytic anemia. Haemolytic anemia is caused by either acquired or hereditary cause. Immunohaemolytic, microangiopathic, direct toxic effects by parasites (e.g. malaria), bacteria (e.g. septicemia by clostridium, staphylococci), snake and spider bites, metals (e.g. lead, copper), extensive burns, PNH (paroxysmal nocturnal haemoglobinuria), splenomegaly are the acquired causes. Hereditary spherocytosis, G6PD (glucose 6



phosphate dehydrogenase) deficiency, PK (pyruvate kinase) deficiency, sickle cell syndrome, thalassemia are the hereditary causes of haemolytic anemia.

### Macrocytic anemia

(see Figure 63) In this type, MCV is increased. Alcohol consumption, use of certain drugs like zidovudine, hydroxyurea is associated with macrocytosis. During above said possibilities i.e. alcohol consumption and drug intake. The next step is to rule out nutritional causes (vitamin B12 and folic acid deficiencies). In this step, serum homocysteine and serum methylmalonic acid level should be ordered to distinguish vitamin B12 from folate deficiency. In vitamin B12 deficiency, both are elevated where as in folic acid deficiency, there is only elevation of homocysteine and not of methylmalonic acid. Later on vitamin B12 deficiency and folate deficiencies are confirmed by the evaluation of vitamin B12 assay and folate assay respectively. In patients with vitamin B12 deficiency, the next step is to screen for the presence of intrinsic factor antibodies. If intrinsic factor antibodies are present, then working diagnosis of pernicious anemia (PA) can be made.

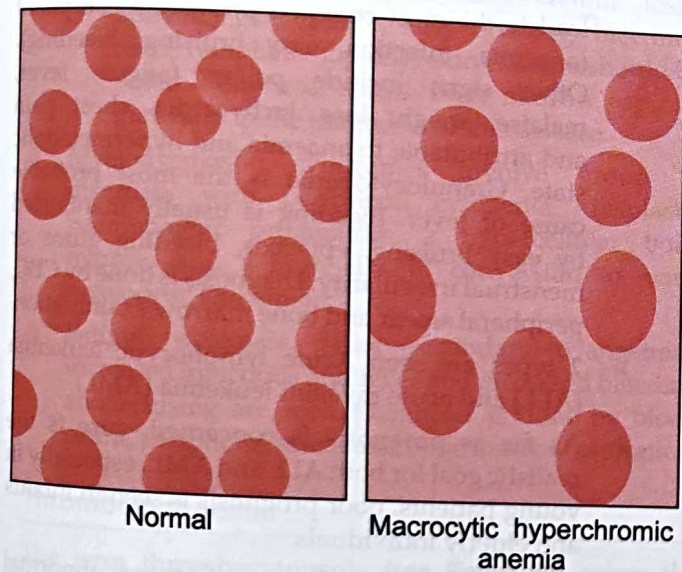


Figure 63: Image of normal and macrocytic hypochromic anemia

### Nutritional Anaemia

The term nutritional anaemia refers to the development of anaemia due to the deficiency of nutritional elements during the formation, maturation of RBCs. Iron and globin, a type of protein are necessary for the formation of haemoglobin. Intrinsic factor, vitamin B12 and folic acid are essential for the maturation of RBCs. Iron deficiency and protein energy malnutrition are responsible for the deficient haemoglobin, whereas the deficiency of intrinsic factor, vitamin B12 and folic acid are responsible for the manifestation of megaloblastic anaemia. Alimentary

canal dysfunction which may be associated with digestion, assimilation of above said factors may lead to the manifestation of nutritional anaemia. The clinical features are similar to the description mentioned in the context of anaemia.

### Thalassemia

If the ferritin level either normal or elevated, clinicians may think of other two possibilities. In thalassemia, RBC count and RDW are either normal or elevated. Polychromasia, target cells, schistocytes, basophilic stipplings may be reported in peripheral blood smear. Here Hb electrophoresis should be done. In thalassemia, we can observe increase in serum iron, transferrin-iron saturation % and decrease in TIBC and serum transferrin. We have to take the advice of haematologist for the accurate diagnosis in this situation.

### Sickle Cell Anaemia

Sickle cell anaemia (see Figure 64) is a type of anaemia observed as a result of chronic haemolysis which occurs due to the shape of RBC as a sickle. Sickle shaped RBCs clog capillaries, causing organ ischemia. Acute exacerbations (crises) may develop frequently. Infection, bone marrow aplasia or lung involvement (acute chest syndrome) can develop acutely and be fatal. Normocytic haemolytic anaemia is characteristic. Diagnosis requires Hb electrophoresis.

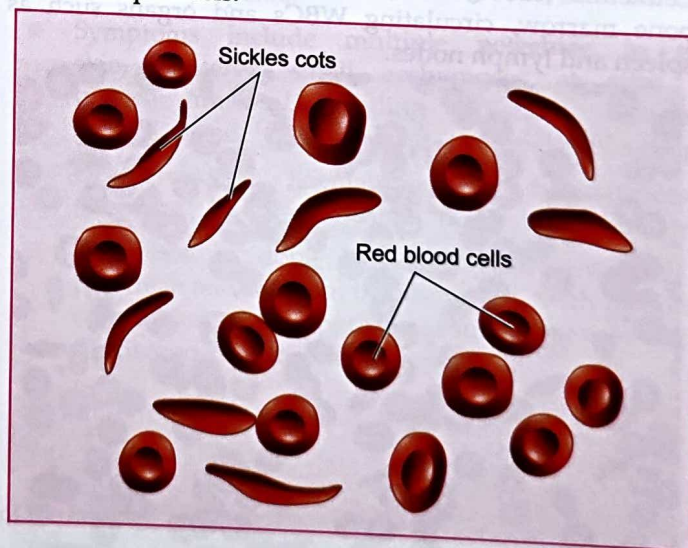


Figure 64: Image of sickle cell anemia

- Due to the shape of RBCs, sickle is the name given to the disease as sickle cell anaemia. In this type of anaemia, there will be distorted, inflexible RBCs adhere to the vascular endothelium and plug small arterioles and capillaries, which leads to infarction. Venous plugging predisposes to thrombosis. Chronic compensatory marrow hyperactivity may lead to deformity of bones.
- Most symptoms occur only in patients who are



homozygous and result from anaemia and vaso occlusive events resulting in tissue ischemia and infarction. Anaemia is usually severe but varies highly among patients, mild jaundice and pallor may be accompanied.

- ⊗ Patients may be poorly developed and often have a relatively short trunk, with long extremities and a tower shaped skull. Hepatosplenomegaly is common in children, cardiomegaly and systolic murmurs are common.
- ⊗ Painful crisis causes severe pain in long bones (e.g. pretibial pain), hands and feet and joints. Joint pain may result from haemarthrosis or avascular necrosis of head of femur. Severe abdominal pain may develop with or without vomiting.
- ⊗ Complications associated with sickle cell anaemia are impaired growth and development. Increased susceptibility to infection, particularly pneumococcal and salmonella infections. Other consequences include ischemic stroke, CNS vasculitis, avascular necrosis of hips, renal concentrating defects, renal failure, heart failure and pulmonary fibrosis.
- ⊗ Diagnosis is usually done by performing peripheral smear test and Hb electrophoresis.

### Leukemia

Leukemias (see Figure 65) are cancers of WBCs involving bone marrow, circulating WBCs and organs such as spleen and lymph nodes.

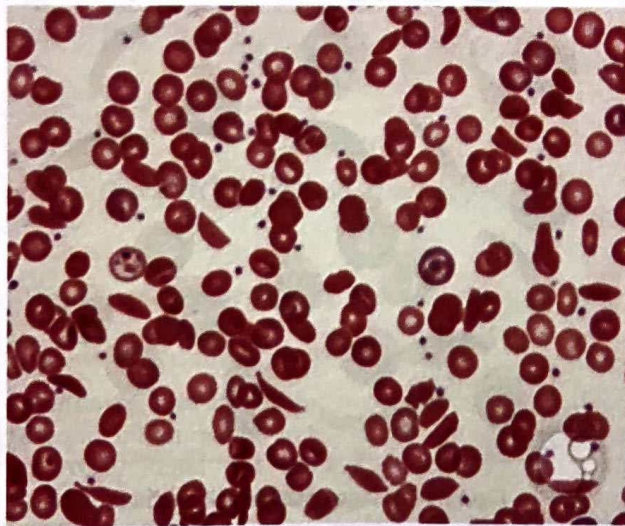


Figure 65: Image of leukemia

- ⊗ The people who are having exposure to ionizing radiation (e.g. post atom bomb in Nagasaki and Hiroshima) or to chemicals such as benzene; chromosomal translocations, preexisting conditions such as immune deficiency disorders, chromosomal disorders, Down syndrome.
- ⊗ Malignant transformation usually occurs at pluripotent stem cell level. Manifestations of

leukemias are due to suppression of normal blood cell formation and organ infiltration by leukemic cells. Anaemia, thrombocytopenia and granulocytopenia may be associated with this disease. Enlargement of liver, spleen lymph nodes may also be present.

- ⊗ Leukemias are classified based on life expectancy as acute or chronic, but now even classification are done according to cellular maturity.

- Acute leukemias
- Chronic leukemias
- Myelodysplastic syndromes

- ⊗ Acute leukemia occurs when a haematopoietic stem cell undergoes malignant transformation into a primitive, undifferentiated cell with abnormal longevity. These lymphocytes (ALL – acute lymphocytic leukemia) or myeloid cells (acute myeloid leukemia) proliferate abnormally, inducing anaemia, thrombocytopenia and granulocytopenia. As the disease is blood borne, this disease can infiltrate various organs and sites including liver, spleen, lymph nodes, CNS, kidneys and gonads.

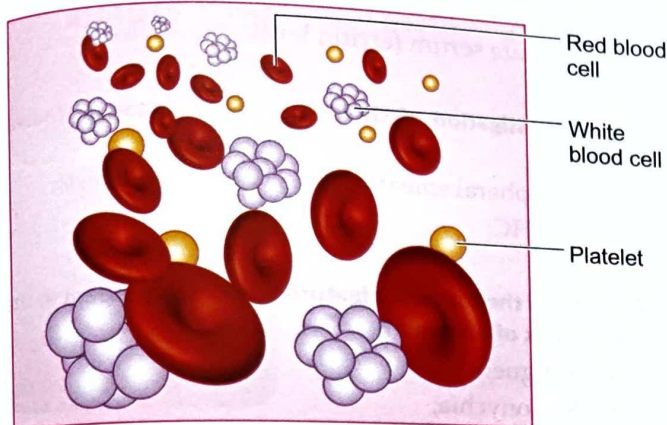
- Symptoms usually present only a few days to weeks before diagnosis. Disrupted RBCs may lead to the most common presenting complaint (anaemia, infection, easy bruising, bleeding). Other signs include pallor, fatigue, fever, malaise, weight loss, tachycardia, chest pain and attributable to anaemia and hypermetabolic state. Granulocytopenia is the most probable cause of fever. Bleeding is usually manifested by easy bruising, epistaxis, bleeding gums or menstrual irregularity. Diagnosis is done by CBC, peripheral smear and bone marrow examination.
- 2 types are seen Acute lymphocytic leukemia (ALL) and acute myeloid leukemia (AML)
- As far as prognosis is concerned, cure is the realistic goal for both ALL and AML, especially in young patients, poor prognosis is seen in infants and elderly individuals.

- ⊗ Chronic leukemia usually manifests as abnormal leukocytosis with or without cytopenia in an otherwise asymptomatic person. Two types of the diseases are seen chronic lymphocytic leukemia (CLL) and chronic myelocytic leukemia (CML).

- CLL is most common type of leukemia where mature appearing defective neoplastic lymphocytes with an abnormally long life span. The peripheral blood, bone marrow, spleen and lymph nodes undergo leukemic infiltration. Symptoms may be absent or may include lymphadenopathy, splenomegaly, hepatomegaly and non specific symptoms attributable to anaemia. Diagnosis is done by peripheral smear examination and bone marrow aspirate.



## Normal Blood



## Platelet Deficiency

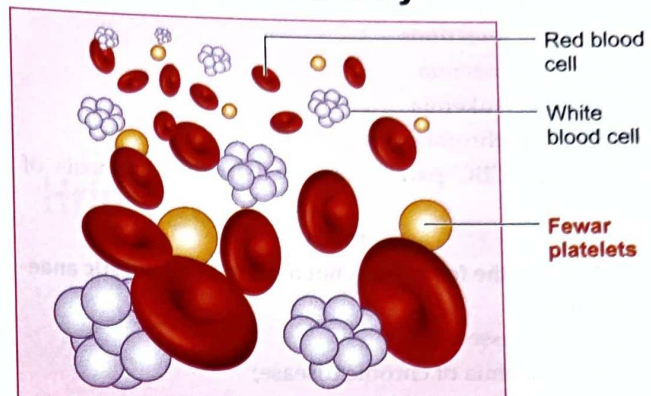


Figure 66: Image of normal blood and blood of thrombocytopenia

- CML occurs when a pluripotent stem cell undergoes malignant transformation and clonal myeloproliferation, leading to a striking overproduction of immature granulocytes. Initially asymptomatic, CML progression is insidious associated with a nonspecific stage (malaise, anorexia and weight loss). Splenomegaly, pallor, easy bruising, bleeding, fever, lymphadenopathy, skin changes should be noted. Peripheral smear, bone marrow aspirate may help in the diagnosis of this disease.
- ⊗ Myelodysplastic syndromes involve group of disorders with peripheral cytopenia, dysplastic haemopoietic progenitors, a hypercellular bone marrow and a high risk of conversion to acute myelocytic leukemia.
- ⊗ Fatigue, weakness, pallor (secondary to anaemia), increased infections and fever and increased bleeding and bruising are seen. Diagnosis is done by blood count, peripheral smear and bone marrow aspiration.

blood. Normal count of thrombocytes is 140,000 to 440,000/ $\mu$ l. These are cell fragments that function in clotting system.

Platelets circulate in the blood for 7 to 10 days. These are destroyed by spleen. When the platelet count  $<50,000/\mu$ l may lead to bleeding tendencies.

- ⊗ Failed platelet production, increased splenic sequestration of platelets, destruction or consumption, dilution of platelets and combination of above mechanisms will be responsible for thrombocytopenia.
- ⊗ Symptoms include multiple petechiae in the skin, scattered small ecchymoses at sites of trauma; mucosal bleeding (epistaxis, bleeding in GI and GU tracts, vaginal bleeding).
- ⊗ Deep visceral haematomas or haemarthroses (bleeding in the joints) may be associated.
- ⊗ Diagnosis is done by clinical presentation of petechiae and mucosal bleeding, CBC with platelets, coagulation studies, peripheral smear test may aid in the diagnosis of thrombocytopenia.

## Thrombocytopenia

The term thrombocytopenia (see Figure 66) refers the decreased number of thrombocytes (platelets) in the

### Key summary of the chapter

- ⊗ Haemopoietic disorders comprise of spectrum of the diseases where in formed elements of the blood from pluripotent haemopoietic stem cell to mature blood cells' dysfunctions are observed. In this chapter, brief description of anaemia, nutritional anaemia, thalassemia, sickle cell disease, leukemia and thrombocytopenia are being discussed with images.
- ⊗ In total, the patient may present with breathlessness which can be observed at any point of time irrespective of the position of the body unlike congestive cardiac failure etc; weakness, fever, palpitation, easy bruising and bleeding, fatigue, scanty menstruation in females of reproductive age group are observed. The person may have pallor, hepatosplenomegaly, haemic murmur (systolic murmur in pulmonary area) etc are seen.
- ⊗ When one makes the diagnosis of anaemia, CBC (Complete blood count) and peripheral smear study has to be done to predict the cause, prognosis of anaemia.



## Self-Assessment: Questions for Reviews

## Short essay questions

- ⊗ Explain anaemia
- ⊗ Explain leukemia
- ⊗ Describe thrombocytopenia
- ⊗ Explain CBC parameters used in the diagnosis of anaemia

## MCQs

1. Which of the following is not a cause of microcytic anaemia?

- a. Thalassemia;
- b. Anaemia of chronic disease;
- c. Iron deficiency anaemia;
- d. Pancytopenia

Answer: (d)

2. Pica, a clinical presentation for iron deficiency anaemia is

- a. Itchiness;
- b. Erectile dysfunction;
- c. Desire to eat weird things;
- d. A small woodland creature

Answer: (c)

3. Haemic murmur seen in the patients of anaemia seen in which part of the chest?

- a. Aortic area;    b. Pulmonary area;
- c. Tricuspid area; d. Mitral area

Answer: (b)

4. The term koilonychia refers to which part of the body?

- a. Eye;                      b. Oral mucosa;
- c. Skin;                    d. Nail

Answer: (d)

5. Which of the following laboratory parameter is used to classify anaemia into microcytic, normocytic or macrocytic?

- a. Mean corpuscular volume (MCV);
- b. Mean corpuscular haemoglobin (MCH);
- c. Mean corpuscular haemoglobin concentration (MCHC);
- d. Red cell Distribution Width (RDW)

Answer: (a)

6. Identify the pathological condition associated with macrocytic anaemia

- a. Sideroblastic anaemia;
- b. Folic acid deficiency;
- c. Autoimmune haemolytic anaemia;
- d. Myelofibrosis

Answer: (b)

7. Pernicious anaemia is an autoimmune disorder that causes

- a. Haemolysis;

- b. Folate excess;
- c. Vitamin B12 deficiency;
- d. Elevate serum ferritin level

Answer: (c)

8. The investigation of choice for the diagnosis of thalassemia is

- a. Peripheral smear;    b. Hb electrophoresis;
- c. MCHC;                      d. RDW

Answer: (b)

9. Which of the following feature is invariably absent in the diagnosis of anaemia?

- a. Fatigue;
- b. Koilonychia;
- c. Scanty or absent menstruation in females;
- d. Icterus

Answer: (d)

10. The commonest cause of jaundice in the patient of thalassemia is

- a. Viral hepatitis C;
- b. Iron deposition in liver;
- c. Viral hepatitis B;
- d. Haemolysis

Answer: (d)

## Chapter resources

- ⊗ <https://www.alamy.com/stock-image-the-effect-of-anemia-on-skin-blood-flow-in-human-164516755.html?imageid=9F2C169F-72FD-4B4B-947E-AC44BA5C2A58&p=298858&pn=1&searchId=f06f476c892e4f52ba284aac2f34cc5b&searchtype=0>
- ⊗ The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- ⊗ Textbook on Clinical biochemistry and haematology with clinical concepts by Naveen Chandra N H, 1st edition, published by the author, 2013.
- ⊗ [www.boldskey.com](http://www.boldskey.com) for microcytic anaemia
- ⊗ [www.researchgate.net](http://www.researchgate.net) for normocytic anaemia
- ⊗ [www.labmedica.com](http://www.labmedica.com) for macrocytic anaemia
- ⊗ Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- ⊗ Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- ⊗ Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- ⊗ Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006
- ⊗ [www.mayoclinic.org](http://www.mayoclinic.org) for sickle cell anaemia
- ⊗ [www.imagebank.hematology.org](http://www.imagebank.hematology.org) for peripheral smear of sickle cell anaemia
- ⊗ [www.newsmedical.com](http://www.newsmedical.com) for thrombocytopenia
- ⊗ [www.wjoanemia.com](http://www.wjoanemia.com)